YAKEEN

NEET 2.0 2027

Physics | Lecture - 2

Units and Measurement

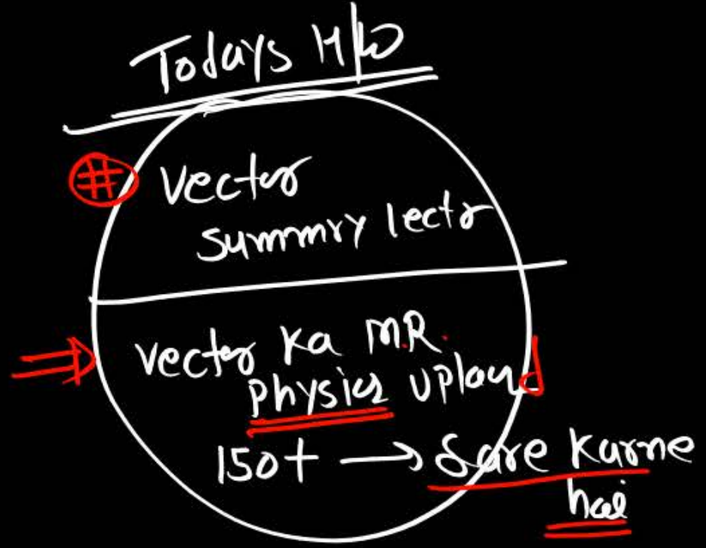
By- Manish Raj (MR Sir)



Todays Goal

Unit & Unit Conversion.

PhD on dimension ✓✓



Unit → Unit is a reference value in-terms of measurment is written.

Mass = 50kg = 50 × [1kg] = 50 × 1000 grm
= 5 × 10^4 grm

Ex

Length = 1m = 100m

L_3 = 5(1m) = 5m = 500cm

Measurent = $n\,u$ = cost^n

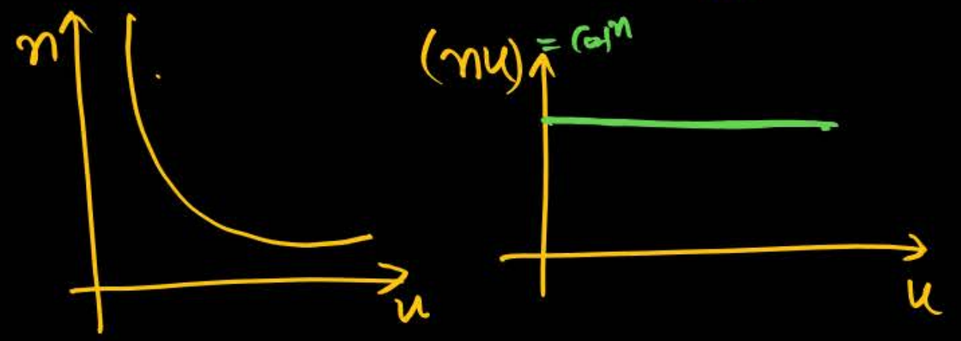
n → magnitude of P.Q

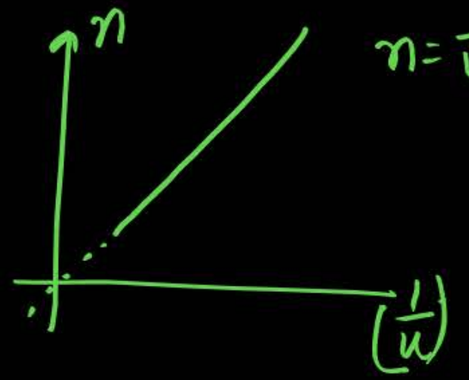
u → Unit of that P.Q.

nu = cost^n

$n_1u_1 = n_2u_2$

nu = cost^n $\quad n \propto \frac{1}{u}$





$n = \frac{\text{cost}^n}{u}$

	SI	cgs unit
mass →	kg	grm
Length	m	cm
Time	sec	sec
Speed/Velocity	m/sec	cm/sec
acceleration	m/s^2	cm/s^2
force $F = ma$	$kg\,m/s^2 = 1\,N$	$grm\,cm/s^2 = 1$ Dyne
Energy	$kg\,m^2/s^2 = 1\,J$	$grm\,cm^2/s^2 = 1$ erg
Power	$kg\,m^2/s^3$	$grm\,cm^2/s^3$
Momentum $P = mV$	$kg\,m/s$	$grm\,cm/sec$

Area → m^2 cm^2

Volume → m^3 cm^3

density → kg/m^3 grm/cm^3

Question COPY

If area of object is $5m^2$ then find its value in CGS unit?

M-1 using $n_1u_1 = n_2u_2$ ✓

M-2

$$\text{Area} = 5\,m^2$$

$$= 5\,[1m]^2$$

$$= 5\,[100cm]^2$$

$$\text{Area} = 5 \times 10^4\ cm^2$$

Question

(2)

Convert 25 m/sec in CGS unit?

Speed = 25 m/sec

$= 25 \left[\frac{100\,cm}{sec}\right]$

$= 2500$ cm/sec

COPY

③ Convert 36 km/hr into m/s.

Sol^{n}

$$\text{Speed} = 36 \text{ km/hr}$$

$$= \cancel{36} \times \frac{1000 \text{ m}}{\cancel{60} \times \cancel{60} \text{ s}}$$

$$= 10 \text{ m/s}$$ ✓

$1 \text{ km} = 1000 \text{ m}$ ✓

$1 \text{ hr} = 60 \text{ mint} = 60 \times 60 \text{ sec}$

$$\text{Speed} = x \frac{\text{km}}{\text{hr}} = x \times \frac{5}{18} \text{ m/s}$$ ✓

Question

④ Convert 1 Newton into Dyne.

$$\text{Force} = 1N = 1\ \frac{kgm}{s^2}$$

$$= \frac{1\times10^3 gm \times 10^2 cm}{s^2}$$

$$= 10^5 \text{ Dyne}$$

$n_1u_1 = n_2u_2$

↓

6-7 line

Question

Convert values of 10 Joule in CGS unit?

$$\text{Energy} = 10 \text{ kg m}^2/\text{s}^2$$

$$= 10 \frac{10^3 \text{g} \times 10^4 \text{cm}^2}{\text{s}^2}$$

$$= 10 \times 10^7 \text{ erg}$$

⑥ If 1m new system of unit of length is 3m then find value of $5m^2$ Area in this system of unit ✓

Soln

S.I = 1m

dash unit = 1m'

$1m' = 3m$

$$\text{Area} = 5m^2$$

$$= 5 \times \left[\frac{1m'}{3}\right]^2$$

$$= \left(\frac{5}{9}\right) m'^2$$

m'^2 → Area ki new Unit

$m' = (\text{dot})$

Question

If unit of length 10m and unit of mass is 5 kg and unit of time is 2 sec then, find value of 10 Joule energy is new system of unit.

dash syste of un.

$1m' = 10m$

$1kg' = 5kg$

$1s' = 2s$

$$\text{Enegy} = 10\left(\frac{kgm^2}{s^2}\right)$$

$$= 10\ \frac{1kg'}{5}\ \frac{1m'^2}{100} \times \frac{1s'^2}{4}$$

$$= \frac{4}{50}\ \frac{kg'm'^2}{s'^2}$$

$$= \frac{2}{25}$$

Question

(8) If unit of length in new system is 2m and unit of mass is 10 kg and unit of time 4 sec then find value of 15 N force in new system.

$1m' = 2m$
$1kg' = 10kg$
$1s' = 4sec$

$$\text{force} = 15 \frac{kg\,m}{s^2}$$

$$= 15 \left[\frac{1kg'}{10} \times \frac{1m'}{2} \times \frac{1}{\frac{1s'}{16}} \right]$$

$$= 15 \times \frac{16}{20} \frac{kg'm'}{s'^2}$$

$$= 12 \frac{kg'm'}{(s'^2)}$$

Question

Q9

In new system of unit, length is x m and mass is y kg, then value of density 5 kg/m^3 in new system will be?

$1m' = x\,m$

$1kg' = y\,kg$

$$\text{density} = 5\left[\frac{kg}{m^3}\right]$$

$$= 5 \times \frac{kg'}{y} \times \frac{}{\left(\frac{1m'}{x}\right)^3}$$

$$= \frac{5x^3}{y}\frac{kg'}{m'^3}$$

Question

If unit of length 10m and unit of mass is 5 kg and unit of time is 2 sec then, find unit of energy in new system.

dash sys.

$1\,kg' = (5\,kg)$

$1m' = (10m)$

$1s' = 2\,sec$

In dash system unit of energy is

$$= \left[\frac{1\,kg'\,(1m')^2}{(1s')^2}\right] = \frac{5\,kg \times (10m)^2}{(2s)^2} = \frac{5\times 100}{4}\,\frac{kgm^2}{s^2}$$

$$= 125\,\left(\frac{kgm^2}{s^2}\right)$$

$1J' = 125\,J$

copy at ghar

MR* Box

P.Q ka value S.I unit me given hai ⟶ New/C.G.S unit me value puchha hai ✓

S.I unit me value likho then S.I unit ko C.G.S/new unit me convert karo.

MR* Box

P.Q ka (new unit) ka value S.I unit ke terms me nikalna ho →

P.Q ka new unit likho then new unit ko ⟶ S.I me convert karo

Question

(9)

If unit of length in new system is 2m and unit of mass is 10 kg and unit of time 4 sec then unit of force in new system.

$1m' = 2m$

$1kg' = 10kg$

$1s' = 4sec$

$$\text{Force} = \frac{1kg'\,m'}{(1s')^2}$$

$$= \frac{10kg \times 2m}{(4s)^2}$$

$$= \frac{20}{16}\,\frac{kgm}{s^2}$$

$$= \frac{5}{4}\,\frac{kgm}{s^2}$$

Question

In new system of unit length is x m and mass is y kg, then unit of density in new system.

Question

M/o

If unit of length becomes double than S.I unit of length, then area $5m^2$ in new unit will be?

DIMENSION LESS PHYSICAL QUANTITY

- Angle/ Solid angle/ Strain $= \frac{\Delta l}{l}$ (θ) [Ω]

  Poisson's ratio, refractive index $\left[\mu = \frac{c}{v}\right]$

  Trigonometry formula/ exponential functions, relative permittivity, efficiency, ratio, pure no. specific gravity (relative density)

$$\left[\varepsilon_r = \frac{\varepsilon_m}{\varepsilon_0}\right]$$

$$\left[\mu_r = \frac{\mu_m}{\mu_0}\right]$$

$$\eta = \frac{\text{output Power}}{\text{Input Power}}$$

$$RD = \frac{\rho_l}{\rho_w} \quad \#$$

$$\sin\theta = \frac{P}{H}$$

$$y = [e^x]$$

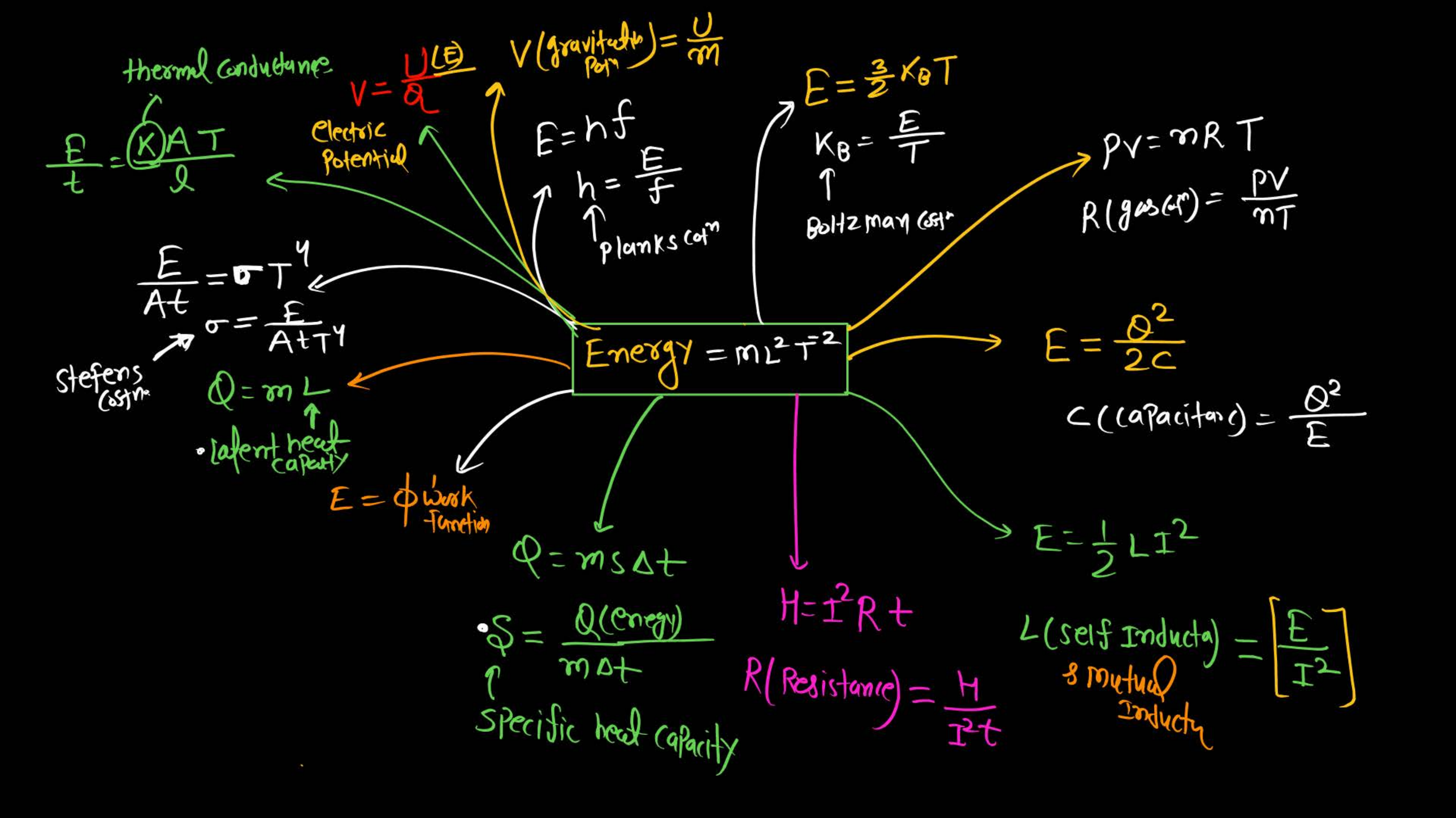
(2) Find dimension of relative velocity ??

$$\vec{V}_{12} = \vec{V}_1 - \vec{V}_2$$

$\vec{V}_{12}$ → m/s

$$\text{Dim}^n = LT^{-1}$$

- Gradient → Change per unit length
- Velocity gradient
- Temperature gradient
- Length gradient
- Time gradient


Energy = mL^2T^{-2}
$V = \frac{U(E)}{q}$
Electric Potential
$V(\text{gravitational Pot}^n) = \frac{U}{m}$
$E = hf$
$h = \frac{E}{f}$
planks cotn
$E = \frac{3}{2} K_B T$
$K_B = \frac{E}{T}$
Boltzmann Cotn
$PV = nRT$
$R(\text{gas cotn}) = \frac{PV}{nT}$
thermal conductance
$\frac{E}{t} = \frac{KAT}{l}$
$\frac{E}{At} = \sigma T^4$
$\sigma = \frac{E}{AtT^4}$
Stefens Cotn
$Q = mL$
• Latent heat capacity
$E = \phi$ work function
$E = \frac{Q^2}{2C}$
$C(\text{capacitance}) = \frac{Q^2}{E}$
$Q = ms\Delta t$
$s = \frac{Q(\text{energy})}{m\Delta t}$
Specific heat capacity
$H = I^2Rt$
$R(\text{Resistance}) = \frac{H}{I^2t}$
$E = \frac{1}{2} LI^2$
$L(\text{self Inducta}) = \left[\frac{E}{I^2}\right]$
& mutual Inducta

copy

① Plank Costn

$$E = hf$$

$$h = \frac{E}{f} = \frac{ML^2T^{-2}}{T^{-1}}$$

$$h = ML^2T^{-1}$$

② Boltzman Cotn

$$E = \frac{3}{2} K_B T$$

$$ML^2T^{-2} = K_B K$$ (K: Kelvin)

$$K_B = ML^2T^{-2}K^{-1}$$

③ $$E = \frac{1}{2} L I^2$$

$$L = \frac{E}{I^2} = ML^2T^{-2}A^{-2}$$

④ $$E = \frac{Q^2}{2C}$$

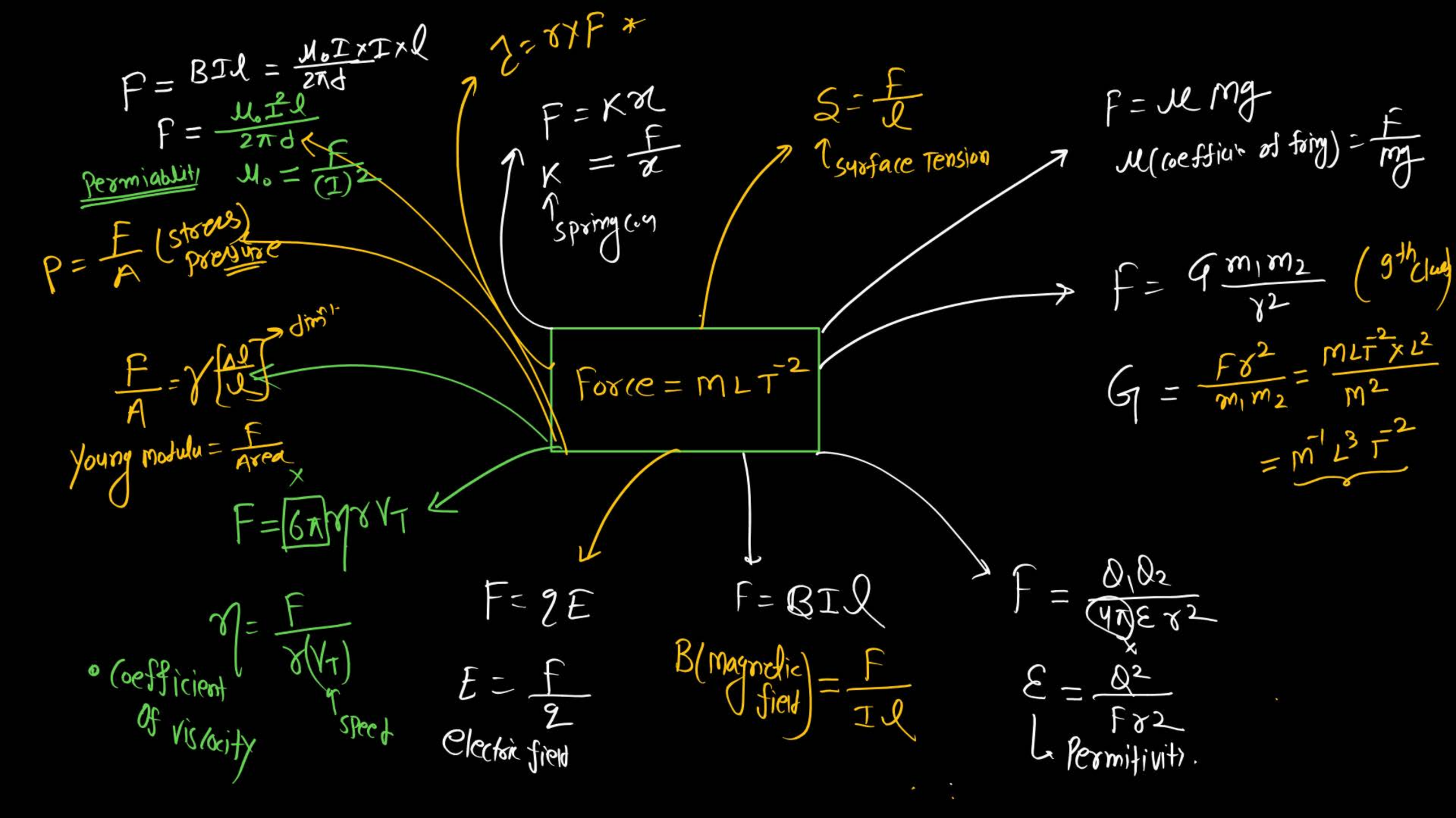
$$C = \frac{Q^2}{E} = \frac{A^2T^2}{ML^2T^{-2}}$$

$$= M^{-1}L^{-2}A^2T^4$$

CHARGE

- **Electric Potential →** $V = \frac{U}{q}$

- **Gravitational Potential →** $V = \frac{\theta}{m}$


$F = BIl = \frac{\mu_0 I \times I \times l}{2\pi d}$
$F = \frac{\mu_0 I^2 l}{2\pi d}$
Permiability $\mu_0 = \frac{F}{(I)^2}$
$\tau = r \times F$ *
$F = Kx$
$K = \frac{F}{x}$
↑ spring const
$S = \frac{F}{l}$
↑ surface Tension
$F = \mu mg$
μ (coefficient of friction) $= \frac{F}{mg}$
$P = \frac{F}{A}$ (stress) pressure
$\frac{F}{A} = Y\left[\frac{\Delta l}{l}\right]$ dimn.
Young modulu $= \frac{F}{\text{Area} \times \frac{\Delta l}{l}}$
Force $= MLT^{-2}$
$F = G\frac{m_1 m_2}{r^2}$ (9th Class)
$G = \frac{Fr^2}{m_1 m_2} = \frac{MLT^{-2} \times L^2}{M^2}$
$= M^{-1}L^3T^{-2}$
$F = 6\pi\eta r V_T$
$\eta = \frac{F}{r(V_T)}$
• Coefficient of viscosity
↑ speed
$F = qE$
$E = \frac{F}{q}$
Electric field
$F = BIl$
B (magnetic field) $= \frac{F}{Il}$
$F = \frac{Q_1 Q_2}{4\pi\varepsilon r^2}$
$\varepsilon = \frac{Q^2}{Fr^2}$
↳ Permitivity.

- **Magnetic Flux:**

$F = qE$

$\phi = E \times \text{Area}$

- **Electric Flux:**

$F = BIl$

$\phi = BA$

- Angular momentum and Plank's constant have same dimensional formula **$[ML^2L^{-1}]$**
- Electric field and potential gradient have same dimensional formula **$[MLT^{-3}\ A^{-1}]$**
- Surface tension, force gradient and spring constant have same dimensional formula **$[ML^0\ T^{-2}]$**
- Acceleration and gravitational field intensity have same dimensional formula **$[M^0\ LT^{-2}]$**

Wins Cotn

$$\lambda = \frac{b}{T}$$

$$b = \lambda T \leftarrow \text{Temp}$$

$$b = m^0 L^1 K^1$$

Question

The dimension of mutual inductance is: [JEE Main 2022]

1. $[ML^2 T^{-2} A^{-1}]$
2. $[ML^2 T^{-3} A^{-1}]$
3. $[ML^2 T^{-2} A^{-2}]$
4. $[ML^2 T^{-3} A^{-2}]$

$E = \frac{1}{2} L I^2$

$L = \frac{E}{I^2} = mL^2 T^{-2} A^{-2}$

Question

The SI unit of a physical quantity is pascal-second. The dimensional formula of this quantity will be

[JEE Main 2022]

1. $[ML^{-1}T^{-1}]$
2. $[ML^{-1}T^{-2}]$
3. $[ML^{2}T^{-1}]$
4. $[M^{-1}L^{3}T^{0}]$

Pressure × time

$= \frac{F}{A} \times T$

$= \frac{MLT^{-2}}{L^2} \times 1$

$= ML^{-1}T^{-1}$

Energy density $= \frac{\text{Energy}}{\text{Volum}}$

$= \frac{ML^2T^{-2}}{L^3}$

$= ML^{-1}T^{-2}$

No copy only Practices

Question

Match List-I with List-II:

[JEE Main 2023]

List-I		List-II	
A.	Planck's constant (h)	I.	$[M^1 L^2 T^{-2}]$
B.	Stopping potential (Vs)	II.	$[M^1 L^1 T^{-1}]$
C.	Work function (ϕ)	III.	$[M^1 L^2 T^{-1}]$
D.	Momentum (p)	IV.	$[M^1 L^2 T^{-3} A^{-1}]$

1. A-III, B-I, C-II, D-IV
2. A-III, B-IV, C-I, D-II
3. A-II, B-IV, C-III, D-I
4. A-I, B-III, C-IV, D-II

Question

Match List-I with List-II:

Choose the correct answer from the options given below: **[JEE Main 2023]**

List-I		List-II	
A.	Surface tension	I.	$Kg\ m^{-1}\ s^{-1}$
B.	Pressure	II.	$Kg\ m^{-1}$
C.	Viscosity	III.	$Kg\ m^{-1}\ s^{-2}$
D.	Impulse (momentum)	IV.	$Kg\ s^{-2}$

1. A-IV, B-III, C-II, D-I
2. A-IV, B-III, C-I, D-II
3. A-III, B-IV, C-I, D-II
4. A-II, B-I, C-III, D-IV

Question

Match List-I with List-II:

Choose the correct answer from the options given below: **[JEE Main 2023]**

	List-I		List-II
A.	Young's Modulus (Y)	I.	$[M\ L^{-1}\ T^{-1}]$
B.	Co-efficient of Viscosity (η)	II.	$[M\ L^{2}\ T^{-1}]$
C.	Planck's Constant (h)	III.	$[M\ L^{-1}\ T^{-2}]$
D.	Work Function (ϕ)	IV.	$[M\ L^{2}\ T^{-2}]$

1. A-II, B-III, C-IV, D-I
2. A-III, B-I, C-II, D-IV
3. A-I, B-III, C-IV, D-II
4. A-I, B-II, C-III, D-IV

Question

Match List-I with List-II:

Choose the correct answer from the options given below: **[JEE Main 2023]**

1. A-III, B-II, C-I, D-IV
2. A-II, B-III, C-IV, D-I
3. A-III, B-II, C-IV, D-I
4. A-II, B-III, C-I, D-IV

List-I (Quantity)		List-II (Dimensional Formula)	
A.	Pressure gradient	I.	$[M^0 L^2 T^{-1}]$
B.	Energy density	II.	$[M^{-1} L^{-1} T^{-2}]$
C.	Electric field	III.	$[M^1 L^{-2} T^{-2}]$
D.	Latent heat	IV.	$[M^1 L^1 T^{-3} A^{-1}]$

Question

Match List-I with List-II:

Choose the correct answer from the options given below: **[JEE Main 2023]**

List-I		List-II	
A.	Angular momentum	I.	$[ML^2 T^{-2}]$
B.	Torque	II.	$[ML^{-2} T^{-2}]$
C.	Stress	III.	$[ML^2 T^{-1}]$
D.	Pressure gradient	IV.	$[ML^{-1} T^{-2}]$

1. A-I, B-IV, C-III, D-II
2. A-III, B-I, C-IV, D-II
3. A-II, B-III, C-IV, D-I
4. A-IV, B-II, C-I, D-III

Question

Copy

Match List-I with List-II:

Choose the correct answer from the options given below: **[JEE Main 2023]**

List-I		List-II	
A.	Torque	I.	Nms^{-1}
B.	Stress	II.	$J\ kg^{-1}$
C.	Latent Heat	III.	Nm^{-1}
D.	Power	IV.	Nm^{-2}

1. A-III, B-II, C-I, D-IV
2. A-III, B-IV, C-II, D-I
3. A-IV, B-I, C-III, D-II
4. A-II, B-III, C-I, D-IV

Question

The pair of quantities having same dimensions is (Kar. NEET 2013)

1. Impulse and Surface Tension
2. Angular momentum and Work
3. Work and Torque
4. Young's modulus and Energy

- **Dimension of electric resistance** **(2007)**

$H = I^2 R t$

- **Self-Induction** **(1989)**

$E = \frac{1}{2} L i^2$

- **Permeability** **(1991)**

- **Magnetic field/Magnetic flux** **(1999)**

$$F = BIl$$

- **Coefficient of viscosity**

$$F = 6\pi\eta v r$$

Question

If the dimensions of a physical quantity are given by $M^a L^b T^c$, then the physical quantity will be **(2009)**

1. velocity if $a = 1, b = 0, c = -1$
2. acceleration if $a = 1, b = 0, c = -2$
3. force if $a = 0, b = -1, c = -2$
4. pressure if $a = 1, b = -1, c = -2$

$M^a L^b T^c$

Question

Which two of the following five physical parameters have the same dimensions?

1. energy density
2. refractive index
3. dielectric constant
4. Youngs modulus
5. magnetic field

(2008)

1. 1 and 4
2. 1 and 5
3. 2 and 4
4. 3 and 5

Question

Which pair do not have equal dimensions ? (2000)

1. Energy and torque
2. Force and impulse
3. Angular momentum and Planck's constant
4. Elastic modulus and pressure.

Question

The density of a material in CGS system of units is 4 g cm^{-3}. In a system of units in which unit of length is 10 cm and unit of mass is 100 g, the value of density of material will be **[Mains 2011]**

1. 0.04
2. 0.4
3. 40
4. 400


@MRPHYSICSS

Thank

You